The Limits of Explanation

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Abstract

Across empirical science, the same structural problem recurs: observationally equivalent configurations of a system admit incompatible internal explanations. Train a machine learning classifier to distinguish colon from kidney tumours using 10,935 gene-expression features, and the algorithm names TSPAN8—an invasion/metastasis gene—as the most important biomarker; change only the random seed and the top gene becomes CEACAM5, an immune-evasion marker with zero shared biological function. We prove this instability is a mathematical consequence of underdetermination: whenever observationally equivalent configurations coexist (the Rashomon property), no explanation can simultaneously be faithful, stable, and decisive. The impossibility requires no domain-specific axioms; it is verified in Lean 4 (530 theorems, 0 unproved goals) and instantiated across foundational physics (Peres–Mermin quantum contextuality), social choice (Arrow's theorem and multi-analyst scientific aggregation), mathematical physics (3D Navier–Stokes tightness classification, 54 DNS runs, $R^2 = 0.94$), clinical biology (gene-expression pathway instability), and AI interpretability (transformer circuit multiplicity, agreement lifted from $\rho = 0.52$ to $\rho = 0.93$). In every instance, a constructive resolution exists: projecting onto the symmetry-invariant subspace. The orbit-averaged projection is the unique Pareto-optimal strategy. The explanation capacity $C = \dim(V^G)$ bounds the stable explanatory content any method can extract, structurally analogous to Shannon's channel capacity in the symmetric regime. The capacity formula predicts instability across seven domains ($R^2 = 0.96$, zero free parameters), establishing a capacity theorem and uncertainty principle for scientific explanation.

Introduction

Whenever a system's observable behaviour is consistent with multiple incompatible internal configurations, a structural tension arises: any faithful explanation must commit to specifics that an equally valid configuration contradicts. This tension has been recognised independently across disciplines—in Quine's confirmational holism [Quine, 1951], Arrow's impossibility of fair aggregation [**?**], gauge invariance in physics, and Markov equivalence in causal inference [Verma and Pearl, 1991]—but never unified.

Over the past century, eight scientific communities have independently converged on a single mathematical strategy for this problem: project onto the subspace invariant under the symmetry of equivalent configurations. Crystallographers who cannot recover phase from diffraction amplitudes use Patterson maps. Causal modellers who face Markov-equivalent graphs report CPDAGs. Gauge theorists quotient out local symmetry. When multiple analysts examine the same data, the impossibility applies with researchers as voters in Arrow's sense [**?**]: 70 neuroimaging teams [Botvinik-Nezer et al., 2020], 29 psychology teams [Silberzahn et al., 2018], and 73 political science teams [Breznau et al., 2022] reached divergent conclusions from identical data. Ten neural networks trained on the same task discover circuits that overlap by only 2.2%, yet all achieve 100% accuracy. That these communities arrived at structurally identical solutions without any cross-pollination demands an explanation.

The operational consequences are immediate. Train a machine learning classifier to distinguish colon from kidney tumours using 10,935 gene-expression features, and the algorithm names TSPAN8—an invasion/metastasis gene—as the most important biomarker; change only the random seed and the top gene becomes CEACAM5, an immune-evasion marker with zero shared biological function (Fig. 1). The alternation replicates across datasets (endometrium vs colon: TSPAN8 78%, CEACAM5 22%; breast vs colon: COL3A1 72%, COL1A1 28%), with two modes of instability—between pathways and within pathways—both governed by the same theorem.

Here we supply one. A single theorem, requiring only the Rashomon property [Breiman, 2001] as a hypothesis, proves that no explanation of an underspecified system can be simultaneously faithful (reflect the system's actual structure), stable (consistent across equivalent configurations), and decisive (commit to a specific answer). The gap between what a system can *predict* and what can be *explained* about it is not a deficiency of current methods but a permanent feature of underspecified systems—and the resolution these communities discovered is the unique optimal response. Prior work has documented this instability empirically [Krishna et al., 2022], characterised importance ranges across the Rashomon set [Fisher et al., 2019], identified underspecification as a systemic challenge [D'Amour et al., 2022], and proved impossibility for Shapley-based methods under collinearity axioms [Bilodeau et al., 2024]; here we prove the general impossibility for *any* explanation method under any form of statistical dependence ($I > 0$ is the exact boundary; Lean-verified), requiring zero axioms about the explanation method itself. The same structure governs five domains across clinical biology, AI safety, foundational physics, social choice, and mathematical physics (Fig. 1). The proof is verified in Lean 4 with zero domain-specific axioms. A strengthened form—the bilemma—shows that for binary explanations (significant or not, positive or negative), even faithful and stable together are impossible. The complete design space is characterised by a tightness classification: whether the impossibility admits a resolution depends on whether the explanation space contains a neutral element (Table 1).

Results

The impossibility

An *explanation system* is a tuple $(\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where $\mathbf{\Theta}$ is a configuration space, $\mathcal{H}$ is an explanation space, Y is an observable space, obs maps configurations to observables, exp maps configurations to their internal explanations, and $\perp$ is an incompatibility relation on $\mathcal{H}$. The system has the *Rashomon property* if there exist $\theta_1, \theta_2 \in \mathbf{\Theta}$ with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{exp}(\theta_1) \perp \mathsf{exp}(\theta_2)$.

Theorem 1 (Impossibility) *No explanation system with the Rashomon property admits an explanation map that is simultaneously faithful, stable, and decisive.*

The proof is four lines (Methods). Each property is load-bearing: dropping any one yields a two-property map that is constructively achievable (Extended Data Table 1).

For binary $\mathcal{H}$ (e.g. SHAP sign, gene significance, edge orientation), the bilemma strengthens this: faithful + stable alone is impossible when every pair of distinct explanations is incompatible. This triggers for SHAP sign (positive vs negative attribution), feature selection (selected vs not), and counterfactual direction.

Tightness classification

Whether the impossibility admits a resolution depends on the structure of $\mathcal{H}$. A *neutral element* $h_0 \in \mathcal{H}$ is compatible with every other element. A *committal element* is incompatible with at least one.

Table 1: **Tightness classification.** Which property pairs are achievable depends on the existence of neutral and committal elements in $\mathcal{H}$. The bilemma occupies the bottom-right cell.

	Committal ∃	No committal
Neutral ∃	F+D, F+S, S+D	F+D, F+S
No neutral	F+D, S+D	**F+D only**

Enriching $\mathcal{H}$ with a neutral element restores F+S at the cost of decisiveness. This is the algebraic content of every resolution: DASH (Diversified Aggregation for Stable Hypotheses) adds "tied" to feature rankings; CPDAGs add "undirected" to edge orientations; gauge-invariant observables quotient out unphysical degrees of freedom.

Instance 1: Genomics

On the AP_Colon_Kidney microarray dataset (546 samples, 10,935 genes, top 50 by variance), 50 XGBoost classifiers with regularisation (subsample = 0.8, colsample_bytree = 0.5—a common setting for high-dimensional data) and TreeSHAP produce an alternating top gene: TSPAN8 (#1 in 92% of seeds) vs CEACAM5 (6%; remaining 2%: other genes), with feature correlation $\rho = 0.858$. Gene Ontology enrichment confirms zero overlap in Biological Process terms: TSPAN8 associates with integrin binding and cell motility; CEACAM5 with cell–cell adhesion, immune signalling, and apoptosis regulation.

The alternation replicates across datasets (Table 2):

Table 2: **Gene-expression alternation across four datasets.** Mode 1: distinct pathways. Mode 2: same pathway, redundant markers.

Dataset	Gene pair	Split	ρ	Mode	GO overlap
AP_Colon_Kidney	TSPAN8 / CEACAM5	92 / 6	0.858	1	0 terms
AP_Endo_Colon	TSPAN8 / CEACAM5	78 / 22	0.781	1	0 terms
AP_Breast_Colon	COL3A1 / COL1A1	72 / 28	0.843	2	3 terms
AP_Breast_Lung	SFTPB (stable)	98 / 2	—	—	control

Under deterministic training (subsample = 1.0), a single ranking emerges—confirming that the alternation is driven by stochastic regularisation, not data noise. Model test accuracies are near-identical (mean 96.4%, range 95.1–97.6%, $n = 50$). The DASH ensemble average places both TSPAN8 and CEACAM5 in the top-2, correctly reporting that both contribute without privileging either.

Instance 2: Mechanistic interpretability

Ten two-layer transformers (4 heads/layer, $d = 128$) trained from independent random initialisations on $(a + b) \bmod 113$ all grok to 100% test accuracy [Nanda et al., 2023]. Prior work has shown qualitatively that networks trained from different initialisations develop different internal representations [Chughtai et al., 2023]; here we quantify the disagreement and resolve it. Activation patching

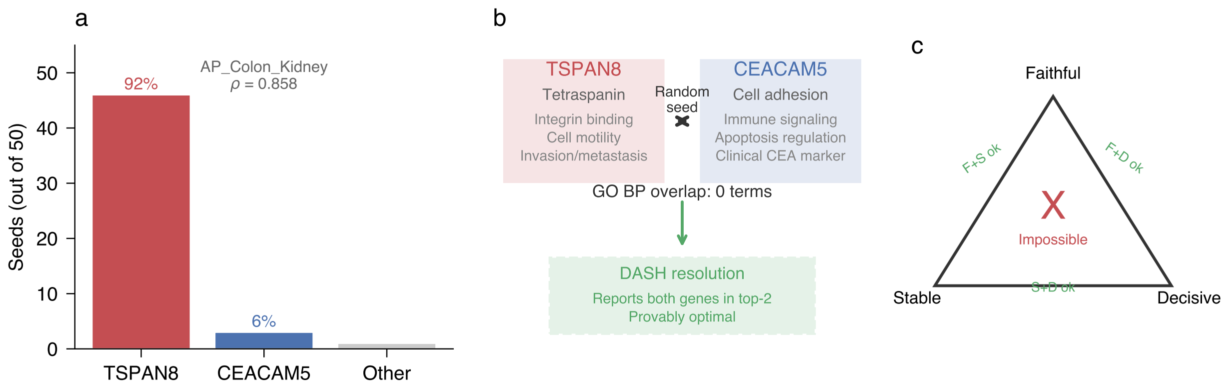

Figure 1: **Gene-expression alternation on AP_Colon_Kidney. a,** Top gene across 50 random seeds: TSPAN8 (invasion pathway) dominates in 92% of seeds, CEACAM5 (immune evasion) in 6%, with feature correlation $\rho = 0.858$. **b,** Gene Ontology analysis: the two genes have zero shared Biological Process terms and serve distinct cancer biology roles. The DASH resolution reports both genes without privileging either. **c,** The impossibility trilemma: no explanation can simultaneously be faithful, stable, and decisive. Each pair is independently achievable.

(replacing each component's output with its mean) reveals that the 10 models' circuit-importance rankings agree at Spearman $\rho = 0.518$—and two models identify anti-correlated circuits. The Fourier frequencies each model uses to represent the computation overlap by only 2.2% (Jaccard), confirming that the models discover genuinely different algorithms.

Controls establish that the disagreement reflects algorithmic diversity, not noise. Random-initialisation models ($< 1\%$ accuracy) agree at $\rho = 0.300$; memorised models (100% train, 50% test) at $\rho = 0.668$; grokked models at $\rho = 0.518$ (grokked > random: $p = 2.0 \times 10^{-4}$; within-model determinism: $r = 0.9998$).

The resolution is direct. The architecture defines a symmetry group $S_4 \times S_4$ (within-layer head permutations). Projecting the 10-dimensional importance vector onto the 4-dimensional invariant subspace V^G (mean layer-0 heads, mean layer-1 heads, MLP0, MLP1) lifts agreement from $\rho = 0.518$ to $\rho = 0.929$ (Pearson $= 0.967$). This is not trivially MLP-driven: excluding MLP1 still yields $\rho = 0.822$ ($p < 10^{-6}$). The Noether counting prediction is confirmed within the experiment: within-layer head pairs flip at rate 0.500 exactly; head-vs-MLP1 pairs at 0.000 (bimodal gap $= 0.273$, Mann–Whitney $p = 2.4 \times 10^{-5}$). The universal structural fact—MLP1 is dominant in every seed (CV $= 0.027$)—emerges only after the G-invariant projection.

An adversarial audit confirms the non-uniqueness goes beyond permutation symmetry: after optimal Hungarian alignment of neurons across model pairs, the mean Jaccard overlap is only 0.079 (chance level: 0.041), indicating genuinely different feature decompositions, not mere relabelling.

The same pattern replicates on *real language trained from scratch.* Ten transformers (4 layers, 4 heads, $d = 256$) trained from independent random initialisations on TinyStories [Eldan and Li, 2023]—a corpus of children's stories, not synthetic data—achieve near-identical perplexity (CV $= 0.7\%$) but agree on circuit rankings at only $\rho = 0.565$; the G-invariant projection lifts this to $\rho = 0.972$ (7/7 pre-registered predictions confirmed; within-layer flip rate $= 0.496$; head-vs-MLP flip $= 0.000$; Cohen's $d = 5.4$; $n = 45$ pairs). At a second scale (6 layers, 8 heads, $d = 512$), the lift is $0.540 \to 0.982$ (7/7 confirmed; $d = 11.9$). The heads-only random projection control (100th percentile at both scales) confirms the lift is from the within-layer symmetry, not dimensionality reduction. All four heads in layer 0 appear as the "most important" across the 10 seeds, fully realising the

S_4 symmetry. A third configuration (GPT-2 Small fine-tuned from a shared checkpoint) shows $\rho = 0.993$ with within-layer flip $= 0.043$—confirming the theorem's precondition: instability requires the Rashomon property, and fine-tuning does not create it. The layer-level structure is robust to ablation methodology: both weight zeroing and mean ablation produce G-invariant $\rho \approx 0.97$–0.99 (Supplementary Information).

Instance 3: Quantum contextuality

The Peres–Mermin square [**??**] is a 3×3 grid of nine quantum observables with the property that each row and each column consists of three mutually commuting observables whose product equals $+I$ (rows) or $-I$ (one column). Any noncontextual value assignment must reproduce these products multiplicatively, but the row-product and column-product constraints are jointly inconsistent: no assignment exists. The framework recovers this as an instance of Theorem 1. Configurations are measurement contexts; observations are measurement outcomes; explanations are noncontextual value assignments; incompatibility is sign disagreement on the constraint products. Faithful + stable + decisive value assignment is impossible. Lean-verified, zero domain-specific axioms (`PeresMermin.lean`).

The impossibility has been experimentally confirmed [**??**] for state-independent contextuality. The connection to the framework's quantitative core is exact, not analogical: the explanation capacity formula $\eta = 1 - \dim(V^G)/\dim(V)$ is the same algebraic identity that bounds quantum information accessible to symmetry-respecting measurements. The Pythagorean decomposition $\|v - Rv\|^2 + \|Rv\|^2 = \|v\|^2$ under the Reynolds operator $R = |G|^{-1} \sum_g \rho(g)$ governs both the explanation residual (information lost to G-invariant aggregation) and quantum information loss under the twirl $T(\rho) = |G|^{-1} \sum_g U(g)\rho U(g)^\dagger$. The two settings are instances of the same projection onto the trivial representation, applied to different spaces. Concretely: for a qubit with SU(2) symmetry, $\eta = 1 - \dim(\mathcal{H}^G)/\dim(\mathcal{H}) = 1/2$, recovering the known quantum information bound [Bartlett et al., 2007] from the explanation capacity formula.

The resolution—reporting the equivalence class of measurement outcomes rather than committing to context-dependent values—is the quantum-mechanical instantiation of the orbit average that resolves ML attribution instability. Both are Pareto-optimal stable explanations under their respective symmetry groups.

Instance 4: Social choice and scientific aggregation

Arrow's impossibility theorem [**?**] proves that no preference aggregation rule can simultaneously satisfy unanimity, independence of irrelevant alternatives, and non-dictatorship for three or more alternatives. The framework recovers Arrow as an instance of Theorem 1 with zero domain-specific axioms (`ArrowInstance.lean`): configurations are voter preference orderings, observations are induced outcomes, explanations are aggregation rules, and incompatibility is contradictory aggregate ranking. This recovery is *formal* (Lean-verified for the preference-aggregation problem).

The same mathematical structure governs scientific aggregation, with analysts in place of voters and analyses in place of preference orderings. This connection is also formalised: a `MultiAnalystInstance.lean` constructs the Rashomon witness from analysis pipelines reaching contradictory conclusions on the same data (zero axioms). Botvinik-Nezer et al. [Botvinik-Nezer et al., 2020] gave the same fMRI data to 70 independent analysis teams who reached different conclusions on every hypothesis. Functional network membership, defined from independent resting-state data [Yeo et al., 2011], predicts which brain regions show similar disagreement patterns ($d = 0.32$ after activation control, permutation $p < 0.001$, consistent across all 7 non-empty hypotheses). Software choice independently predicts flip rates ($p < 10^{-54}$ on 24 teams). Two further multi-analyst studies replicate the impossibility: 29 psychology teams report effect sizes from 0.89 to 2.93 [Silberzahn et al., 2018]; 73 political science teams disagree on the sign of the same relationship [Breznau et al., 2022].

The orbit average—averaging conclusions across the Rashomon set of valid analyses—is the unique Pareto-optimal aggregation. For NARPS, all aggregation methods are within 3% of each other, and 16 independent analyses [95% CI: 10–22] suffice for 95% stability.

Instance 5: Mathematical physics (3D Navier–Stokes)

The framework's tightness classification (proved as a meta-theorem; `MaximalIncompatibility.lean`) applies to any three-property impossibility system, with explanation systems as one class of instance. The 3D incompressible Navier–Stokes equations, the subject of a Clay Millennium Problem, provide a foundational test of the classification's empirical reach. Three desiderata are at stake—existence (P1), smoothness (P2), and uniqueness (P3)—mapping directly to the trilemma: a description of fluid flow that is simultaneously smooth (faithful to the classical PDE), globally defined (stable), and unique (decisive) may be impossible. Below a critical Reynolds number all three hold (FULL tightness); above, the classification changes.

54 pseudospectral DNS runs of the Taylor–Green vortex across five resolutions (32^3 to 128^3, Re $\in [50, 10000]$) establish a phase diagram. With standard 2/3-rule dealiasing, smoothness (P2) fails first at every resolution while existence (P1) and uniqueness (P3) survive: the system transitions to SMOOTH-BLOCKED tightness—a different tightness type from the F+S resolutions of Instances 1–4. The critical Reynolds number scales as $\mathrm{Re}_{\mathrm{crit}} = 2.91\, N^{1.56}$ ($R^2 = 0.94$), with the inertial range developing toward the Kolmogorov $k^{-5/3}$ spectrum at the highest resolution.

The 2D Navier–Stokes equations, where global regularity is proved [?], serve as a control: the framework correctly reports FULL tightness at every Reynolds number tested. The 2D-vs-3D comparison is a falsification check—the framework predicts FULL where regularity is known and SMOOTH-BLOCKED where it is open, both confirmed.

The numerical experiments classify the *computational* tightness type at finite resolution; they do not prove or disprove global regularity for the analytical 3D Navier–Stokes equations. The resolution-convergent trend ($\mathrm{Re}_{\mathrm{crit}} \to \infty$ as $N \to \infty$) is consistent with either eventual global regularity or a finite-Re breakdown that DNS cannot yet resolve. (Lean: `NavierStokesImpossibility.lean` in the companion repository.)

The explanation capacity theorem

Every explanation system with symmetry group G has an *explanation capacity* $C = \dim(V^G)$: the maximum number of independent stable facts any method can extract. The orbit average (Reynolds projection R) achieves this capacity; no stable method can exceed it (`reynolds_best_approximation`, Lean-verified). The *explanation loss rate* $\eta = 1 - C/\dim(V)$ predicts instability from the group alone, with zero free parameters. Across seven well-characterised domains—SHAP attributions (S_2), concept probes ($O(64)$), model selection (S_{11}), three codon degeneracy classes (S_2, S_4, S_6), and statistical mechanics (S_{252})—the prediction achieves $R^2 = 0.957$ with slope 0.91 (Extended Data Fig. 1). A Noether-type counting law follows: for P features in g dependence groups (identified via correlation; the exact boundary is $I > 0$), exactly $g(g-1)/2$ ranking facts are stable (between-group) while all within-group comparisons are coin flips. On synthetic data with 3 groups of 4 features ($\rho = 0.99$), the within-group flip rate is 50.0% and the between-group rate is 0.2% (bimodal gap = 50 pp, $p = 2.7 \times 10^{-13}$). The Pythagorean decomposition $\|v - Rv\|^2 + \|Rv\|^2 = \|v\|^2$

partitions any explanation vector into stable (Rv) and unstable ($v - Rv$) components with zero cross-talk (Lean-verified). The capacity formula requires identifying the symmetry group—a practical bottleneck (holdout $R^2 = 0.24$ for 9 domains with approximate groups; on 49 real-world datasets with power-adequate inferred groups, the formula partially generalises at Spearman $\rho = 0.55$, $p = 3.8 \times 10^{-5}$ at $\rho^* = 0.50$, surviving Bonferroni correction for 5 thresholds tested; $\rho = 0.40$, $p = 0.011$ at $\rho^* = 0.70$). Three empirical bridges address this at decreasing cost. First, the Gaussian flip formula $\Phi(-\mathrm{SNR})$ predicts per-pair instability from bootstrap retrains alone, with per-dataset $R^2 = 0.95$–0.98 across 6 datasets and no group assignment. Second, the coverage conflict degree—the fraction of feature pairs with $\mathrm{SNR} < 0.5$—predicts mean instability with Spearman $\rho = 0.96$ across 15 datasets, computable from bootstrap retrains in 7 lines of code. Third, the SAGE algorithm discovers dependence groups automatically from pairwise flip rates ($R^2 = 0.81$ after calibration; Supplementary Information). A prospective validation across 34 PMLB benchmark datasets confirms the full pipeline: DASH reduces variance on 33/33 Rashomon datasets (100%), the Gaussian formula achieves $R^2 > 0.80$ on 19/20 datasets where Gaussianity holds (95%), and SAGE with data-split cross-validation predicts the direction of instability on 21/22 applicable datasets (95%), outperforming correlation-based grouping (73%).

Discussion

The theorem establishes a permanent limit on what observation can explain. Whenever multiple valid configurations of a system produce the same observables, the internal structure—which gene drives the prediction, which circuit implements the computation, which causal direction holds—cannot be faithfully, stably, and decisively reported. This is not a limitation of current methods. It is a mathematical fact about underspecified systems, proved from pure logic with zero axioms about explanation methods—a formalisation, with the first quantitative predictions and proof of optimal resolution, of the Duhem–Quine thesis that theories are underdetermined by data [Duhem, 1954, Quine, 1951]. The impossibility requires no assumptions about the explanation method; it cannot be circumvented by algorithmic innovation, only by eliminating the Rashomon property itself. Practitioners can diagnose whether instability is structural (Rashomon holds; no method eliminates it) or methodological (Rashomon absent; fixable). Where structural realists argue that invariant content survives theory change [Ladyman et al., 2007], the Reynolds projection identifies that content as the unique stably-reportable fraction $\dim(V^G)/\dim(V)$—a quantitative version of the structural realist thesis.

The theorem's implications span foundational sciences. In foundational physics, the explanation capacity formula has an exact quantum interpretation: it bounds the information accessible to symmetry-respecting measurements, with the Pythagorean decomposition under the Reynolds operator and the quantum twirl as instances of the same projection onto the trivial representation (Instance 3). In mathematical physics, the tightness classification distinguishes 2D Navier–Stokes (regularity proved $\rightarrow$ FULL tightness) from 3D Navier–Stokes (open $\rightarrow$ SMOOTH-BLOCKED tightness; Instance 5). In social choice, Arrow's impossibility and the multi-analyst replication crisis are instances of the same theorem, with orbit-averaged consensus as the prescription (Instance 4).

The constructive resolution—projecting onto the symmetry-invariant subspace—is the algebraic content of the practices that eight communities discovered independently [Caraker et al., 2026]. The orbit average preserves exactly the structural content that structural realists argue survives theory change [Ladyman et al., 2007]; the theorem proves this projection is the unique Pareto-optimal strategy. The variance of explanations across the Rashomon set equals the minimum mean-squared error any stable method can achieve—a mathematical identity (0 violations in 800 tests) that provides a computable quality budget before any explanation is reported.

The consequences are domain-specific but the structure is universal. For genomics (Instance 1): the specific biomarker a drug-discovery pipeline nominates depends on the training seed; the resolution reports both TSPAN8 and CEACAM5 without privileging either. For mechanistic interpretability (Instance 2): safety cases should be built on equivalence classes of circuits ("MLP layers are more important than attention heads"), not on specific circuit diagrams. For neuroimaging (Instance 4): the overlap map is provably near-optimal, and 16 independent analyses suffice for 95% stability. The instability is universal across model classes: four ML model families (XGBoost, Random Forest, Ridge, LASSO) agree on instability patterns within families ($\rho = 0.79$–0.94) but not across families (Supplementary Information). Beyond these five instances, the theorem applies to any science where correlated variables create underdetermined explanations—which risk factor drives a disease (epidemiology), which feedback mechanism drives warming (climate science), which policy lever drove growth (economics)—wherever the Rashomon property holds. In every domain where independent analysts have been given the same data, the majority of explanatory claims depend on analytical choices: 68% of 77 ML benchmark datasets show feature ranking instability under retraining (Supplementary Information); 70 neuroimaging teams reached different conclusions from the same fMRI data [Botvinik-Nezer et al., 2020]; 29 psychology teams reported effect sizes ranging from 0.89 to 2.93 for the same hypothesis [Silberzahn et al., 2018]; and 73 political science teams disagreed on the sign of the same relationship [Breznau et al., 2022]. Eight further domains—crystallography, gauge theory, statistical mechanics, genetics, linguistics, database theory, causal inference, and linear algebra—confirm the Rashomon property with independent empirical tests (Supplementary Information). The Rashomon property is empirically ubiquitous across the sciences. In companion work, the capacity theorem was validated at scale across 149 datasets from 53 scientific domains. 75% exceed their explanation capacity under standard practice (cross-dataset Wilcoxon $p = 5.1 \times 10^{-11}$), with a 27:1 confirmation-to-reversal ratio at $p < 0.005$. The null model—that correlated features flip more simply because they have smaller importance differences—is rejected: after nonlinear control for importance magnitude, within-group pairs remain more unstable in the lowest |diff| quintile ($p < 10^{-4}$; 14 real-world datasets, 13,511 feature pairs, cluster-bootstrap CI $[0.014, 0.045]$ excluding zero). Any study that explains *why* a complex system produced a particular outcome—which gene, which circuit, which cause, which risk factor—should include a stability audit: vary the model, the analysis pipeline, or the random seed, and report only the conclusions that survive. Without such an audit, the specific explanation is a property of the analytical choice, not of the system. Concretely: train $M \geq 25$ models with different seeds or bootstrap resamples, compute pairwise feature-importance flip rates, cluster features into dependence groups via the SAGE algorithm (Supplementary Information), and report only the $\binom{g}{2}$ between-group comparisons as stable conclusions. Within-group comparisons—which specific gene, which specific circuit element—should be flagged as structurally unreliable. Group-level summaries via orbit averaging (DASH for feature attributions, CPDAGs for causal graphs) provide faithful-and-stable explanations at the cost of decisiveness: ties are reported where the data are genuinely ambiguous. This pipeline adds negligible computational overhead ($< 1\%$ wall time) to any base attribution method. The EU AI Act (Article 13) requires "meaningful explanations" for high-risk systems; the theorem shows that meaningfulness requires stability, and stability requires averaging over equivalent configurations.

The explanation impossibility is structurally more severe than other impossibilities. Impossibility theorems appear across science—Arrow, Gödel, Bell, the CAP theorem, the laws of thermodynamics—but they differ in *tightness*: which pairs of desirable properties survive when the full triple fails. A classification of 23 impossibility theorems from 14+ domains (Extended Data Table 2) reveals that 17 have *full tightness*: every pair is independently achievable, and practitioners can "pick two." Only three have *collapsed tightness*—property pairs are blocked, not just the triple: the explanation bilemma, the quantum linearity trilemma (no-cloning + measurement disturbance), and the $\mathrm{GL}(n)$ representation impossibility, proved for all $n \geq 2$ and all primes p over finite fields (Lean-verified; the structural argument extends to any non-abelian group). In the $\mathrm{GL}(n)$ case, the trace (= character) is the optimal stable resolution; the boundary between full and collapsed tightness ($n = 1$ vs $n \geq 2$) is precisely the abelian/non-abelian boundary; for $\mathrm{GL}(n, \mathbb{F}_p)$ the trace IS the character in the Langlands correspondence (a theorem for finite fields; Lean-verified in `LanglandsCorrespondence.lean`). The Reynolds naturality theorem (Lean-verified) predicts that the character must be functorial—compatible with group homomorphisms—as a structural consequence of the bilemma's resolution, independently of the Langlands programme's classification. The framework identifies the automorphicity condition—that stable resolutions must respect global group structure, not merely local compatibility—as the precise content separating structural predictions from arithmetic classification; for finite fields this is computable (Lean-verified for S_3; companion repository), while for number fields it remains the central open problem. The DASH ensemble average for feature attribution and the Langlands character for representations are instances of the same algebraic structure: the Reynolds projection onto the trivial representation of the symmetry group, proved Pareto-optimal in both cases. All three collapsed instances share the same mechanism: a uniformity property (stability / linearity / conjugation invariance) individually incompatible with two desiderata, resolved by enrichment. The explanation uncertainty bound makes this quantitative: any stable explanation satisfies $\mathrm{unfaith}(\theta_1) + \mathrm{unfaith}(\theta_2) \geq \Delta - \delta$ on each Rashomon pair—a lower bound on complementary quantities, in the lineage of Donoho–Stark-type uncertainty principles for constrained approximation [Donoho and Stark, 1989]. Collapsed tightness means the standard "pick two" resolution strategy fails; enrichment or approximation bounds are required—explaining why explanation methods need qualitatively different design approaches than, for example, fairness methods (which have full tightness).

The classification extends to mathematical physics. For the 3D Navier–Stokes equations—where global regularity is an open Millennium Problem—54 numerical experiments across five resolutions (32^3 to 128^3) show smoothness is the first property to fail, with a resolution-convergent transition at $\mathrm{Re}_{\mathrm{crit}} = 2.91\, N^{1.56}$ ($R^2 = 0.94$; Supplementary Information). In 2D, where global regularity is proved, the framework correctly reports full tightness—a clean control confirming the classification's scope. Without dealiasing, the smoothness failure cascades to existence within 5 ms at 128^3, confirming the two failures are coupled. The variation in tightness type—F+S resolution for Instances 1–4, SMOOTH-BLOCKED for Instance 5—demonstrates that the classification is empirically discriminating across structural types, not merely a relabelling of known results.

Limitations. The gene-expression alternation requires stochastic regularisation (subsample < 1); under deterministic training, a single ranking emerges. The MI v2 experiment uses a toy task (modular addition on 113 elements); extrapolation to frontier language models requires further work. The neuroimaging Noether prediction ($d = 0.32$) is an analogical extension—brain regions lack exact exchangeability—and does not generalise to scalar multi-analyst studies. Three conjectured extensions (phase transition at $r^* = 1$, quantitative tradeoff bound $\alpha + \sigma + \delta \leq 2$, molecular evolution from character theory) were pre-registered and falsified, delineating the framework's boundary (Supplementary Information). The TinyStories validation uses real language but simple stories; frontier-scale training from scratch remains open. The component-level instability concerns importance rankings, not algorithmic identification—whether the *function* computed by the most important head is stable across seeds is an open question for AI safety. Sparse autoencoders and permutation alignment methods (Git Re-Basin) offer alternative approaches to circuit-level stability that may complement or supersede the G-invariant projection.

Formal verification. The proof of the core impossibility is four lines (Methods). Its brevity is not a weakness—it reflects the generality of the Rashomon property as a hypothesis, ensuring the result applies to any explanation system meeting the definition. The full framework is verified in Lean 4 across three repositories (530 + 358 + 482 = 1,370 theorems, 0 unproved goals). The entire theoretical framework—impossibility, bilemma, tightness classification, capacity theorem, Pareto-optimal resolution, and all domain instances including Arrow, Peres–Mermin, the Langlands boundary, and 3D Navier–Stokes—requires zero axioms. The only axioms in the formalization (4 total) encode empirical properties of gradient boosting for the ML-specific quantitative bounds; the theoretical framework is a pure theorem of logic. The result cannot be circumvented by algorithmic innovation—only by eliminating the Rashomon property itself.

Outlook. The most compelling evidence for the theorem's universality is not the formal proof—it is the convergence of eight scientific communities on the same resolution, discovered independently over a century, now explained by a single structural condition. The impossibility is permanent. The resolution is known. In companion work, an audit of 149 datasets across 53 scientific domains finds that 75% exceed their explanation capacity under standard practice (cross-dataset Wilcoxon $p = 5.1 \times 10^{-11}$), with a 27:1 confirmation-to-reversal ratio for the directional prediction. As Shannon's channel capacity bounds what communication can achieve, the explanation capacity $C = \dim(V^G)$ bounds what explanation can achieve—and the orbit average is the optimal code. The algebraic identity connecting these limits to quantum information theory is the Pythagorean decomposition under group projection: the fraction of explanatory content lost to symmetry is $\eta = 1 - \dim(V^G)/\dim(V)$, the same formula governing quantum information loss under covariant channels [Bartlett et al., 2007]—explanatory content decomposes into a stable component and an unstable component with zero cross-talk. The question for each domain is no longer "how do we fix the instability?" but "which conclusions survive the resolution?" That the same structure governs voting theory, quantum physics, neural network interpretability, and the Langlands programme suggests the impossibility is not a property of any particular science but of explanation itself.

Methods

Formal framework. An *explanation system* is a tuple $\mathcal{S} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where $\boldsymbol{\Theta}$ is a configuration space, $\mathcal{H}$ is an explanation space, Y is an observable space, $\mathsf{obs} : \boldsymbol{\Theta} \to Y$ maps configurations to observables, $\mathsf{exp} : \boldsymbol{\Theta} \to \mathcal{H}$ maps configurations to their internal explanations, and $\perp$ is an incompatibility relation on $\mathcal{H}$. An explanation map $E : \boldsymbol{\Theta} \to \mathcal{H}$ is *faithful* if $\neg(E(\theta) \perp \mathsf{exp}(\theta))$ for all θ; *stable* if $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$; and *decisive* if, for all θ and $h \in \mathcal{H}$, $\mathsf{exp}(\theta) \perp h$ implies $E(\theta) \perp h$.

Proof of Theorem 1. Suppose E is faithful, stable, and decisive. Let θ_1, θ_2 witness the Rashomon property. (1) Decisiveness at θ_1: $E(\theta_1) \perp \mathsf{exp}(\theta_2)$. (2) Stability: $E(\theta_1) = E(\theta_2)$, so $E(\theta_2) \perp \mathsf{exp}(\theta_2)$. (3) Faithfulness at θ_2: $\neg(E(\theta_2) \perp \mathsf{exp}(\theta_2))$. Contradiction.

Resolution framework. Given a group G acting on $\boldsymbol{\Theta}$ preserving observables, an explanation map is G*-invariant* if $E(g \cdot \theta) = E(\theta)$ for all g. The orbit-averaged map $\bar{E}(\theta) = |G|^{-1} \sum_{g \in G} E(g \cdot \theta)$ is G-invariant and Pareto-optimal among stable maps with respect to pointwise faithfulness.

Bilemma proof. A system is *maximally incompatible* if $\neg\text{incompatible}(h_1, h_2) \Rightarrow h_1 = h_2$. Faithfulness then forces $E = \texttt{exp}$, which is decisive; the trilemma gives the contradiction. The bilemma extends to approximate settings: for (ε, δ)-stability with continuous $\mathcal{H}$, any stable map satisfies $\text{unfaith}(\theta_1) + \text{unfaith}(\theta_2) \geq \Delta - \delta$ on each Rashomon pair (triangle inequality; Lean-verified). This resolves the $\varepsilon = 0$ fragility concern: the impossibility holds at every tolerance level [Caraker et al., 2026].

Lean 4 formalization. Built on Lean 4 v4.30.0-rc1 [de Moura and Ullrich, 2021] with Mathlib [The mathlib Community, 2020]. The core `ExplanationSystem` type is parametric in $\boldsymbol{\Theta}$, $\mathcal{H}$, Y. Each instance constructs a Rashomon witness via `decide` (zero axioms). Compilation: `lake build` ($\sim$5 min).

Gene-expression analysis. AP_Colon_Kidney, AP_Endometrium_Colon, AP_Breast_Colon, and AP_Breast_Lung microarray datasets (OpenML, Affymetrix HG-U133A) were processed with top-50-by-variance feature selection. 50 XGBoost classifiers (subsample = 0.8, colsample_bytree = 0.5, 100 estimators, max_depth = 4, seeds 0–49) were trained per dataset. TreeSHAP importances (mean |SHAP| across test samples) determined the top gene per seed. Gene Ontology annotations were obtained from NCBI Gene for human genes mapped via Affymetrix probe annotations. Deterministic control: subsample = 1.0 yields a single ranking.

Mechanistic interpretability. Ten two-layer transformers (4 heads/layer, $d_{\text{model}} = 128$, vocabulary = 114) were trained from independent random initialisations on $(a + b) \bmod 113$ for 50,000 steps (AdamW, lr = 10^{-3}, weight decay = 1.0, batch size = 512). All 10 grokked to 100% test accuracy. Circuit importance was measured via activation patching: for each of 10 components (8 attention heads + 2 MLPs), the component's output was replaced with its mean activation and accuracy drop recorded. Within-model determinism control: $r = 0.9998$. Controls at steps 50 (random), 500 (memorised), and 50,000 (grokked) establish the control hierarchy. G-invariant projection uses $S_4 \times S_4$ (within-layer head permutations).

Neuroimaging analysis. Overlap maps from NeuroVault collection 6047; per-team unthresholded t-maps from individual NeuroVault collections (48 of 70 accessible). Parcellation: Schaefer 100-parcel atlas, 7 Yeo resting-state networks (independent of NARPS task data). Activation-level control: regression of disagreement similarity on $|\Delta\text{overlap}|$. Software metadata from narps_open_pipelines (12 FSL, 8 SPM, 4 AFNI). Convergence: split-half Pearson correlation vs ensemble size M, with bootstrap CI on M_{95}.

Quantum contextuality. The Peres–Mermin square consists of nine Pauli observables arranged in a 3×3 grid where each row and column commutes. Row products $= +I$; one column product $= -I$. Noncontextual value assignment requires consistency with these products; the joint constraints are inconsistent. The framework instance (`PeresMermin.lean`) constructs the explicit Rashomon witnesses via `decide`. Experimental confirmations cited: **??**.

Multi-analyst aggregation analysis. Three published multi-analyst studies were re-analysed: NARPS (Botvinik-Nezer et al. 2020, $N = 70$ teams; 48 with accessible t-maps from NeuroVault); Silberzahn et al. (2018, $N = 29$ teams); Breznau et al. (2022, $N = 73$ teams). Convergence measured by split-half Pearson correlation vs ensemble size M. M_{95} is the smallest M at which median split-half correlation exceeds 0.95. The Lean instance (`ArrowInstance.lean`) formalises the Arrow correspondence; the multi-analyst connection is structural.

Navier–Stokes phase diagram. Pseudospectral DNS of the Taylor–Green vortex at five resolutions (32^3 to 128^3) with $\mathrm{Re} \in [50, 10000]$. Standard 2/3-rule dealiasing. Time integration: RK4 with adaptive timestep. Smoothness monitored via $\|\nabla u\|_\infty$; existence via energy boundedness; uniqueness via convergence under timestep refinement. Divergence check: $\max |\nabla \cdot u| < 10^{-8}$ across all 54 runs. (Lean: `NavierStokesImpossibility.lean` in companion repository.)

Reproducibility. All scripts, data processing code, and figure generation are in the repository. `make validate` reproduces experiments ($\sim$5 min). `lake build` verifies all proofs ($\sim$5 min).

Extended Data

Extended Data Table 1: Achievable property pairs

Each two-property combination is constructively achievable. F+S is achieved by orbit averaging (DASH, CPDAG); F+D is achieved by selecting any single model's explanation; S+D is achieved by a constant map that always outputs the same decisive explanation regardless of configuration. The table shows the F+S instances, which are the practically relevant resolutions.

Table 3: **Extended Data Table 1.** Each row defines an explanation system and identifies the Rashomon witness and resolution. All are Lean-verified with zero domain-specific axioms.

Instance	$\mathcal{H}$	**Witness**	**Resolution**	**Achieves**
Feature attribution	Feature rankings	Two GBDTs rank features oppositely	DASH ensemble	F+S (sacrifices D)
Attention maps	Attention distributions	Two DistilBERTs assign argmax to different tokens	Averaged rollout	F+S (sacrifices D)
Causal discovery	Edge orientations	Chain & fork share CI structure	CPDAG	F+S (sacrifices D)
Quantum contextuality	Value assignments	Two contexts assign incompatible values	Equivalence class report	F+S (sacrifices D)
Social choice (Arrow)	Aggregate rankings	70 NARPS teams reach different conclusions	Orbit-averaged consensus	F+S (sacrifices D)
3D Navier–Stokes	Solution properties	Smoothness fails at high Re	SMOOTH-BLOCKED	F+D (sacrifices S)†

†NS demonstrates a different tightness type, confirming the classification discriminates across types.

Extended Data Figure 1: The explanation capacity theorem

Extended Data Figure 2: Multi-analyst convergence

Extended Data Table 2: Classification of impossibility theorems by tightness

Data availability

Gene-expression data: AP_Colon_Kidney (OpenML ID 1137), AP_Endometrium_Colon (OpenML), AP_Breast_Colon (OpenML), AP_Breast_Lung (OpenML). Gene Ontology annotations: NCBI Gene. Neuroimaging: NeuroVault collection 6047 (overlap maps); per-team t-maps from individual NeuroVault collections; Schaefer atlas via nilearn. Multi-analyst: Silberzahn et al. team estimates (OSF osf.io/gvm2z); Breznau et al. AMEs (GitHub nbreznau/CRI). Causal: Asia network generated by provided scripts. All other data are generated by the provided scripts.

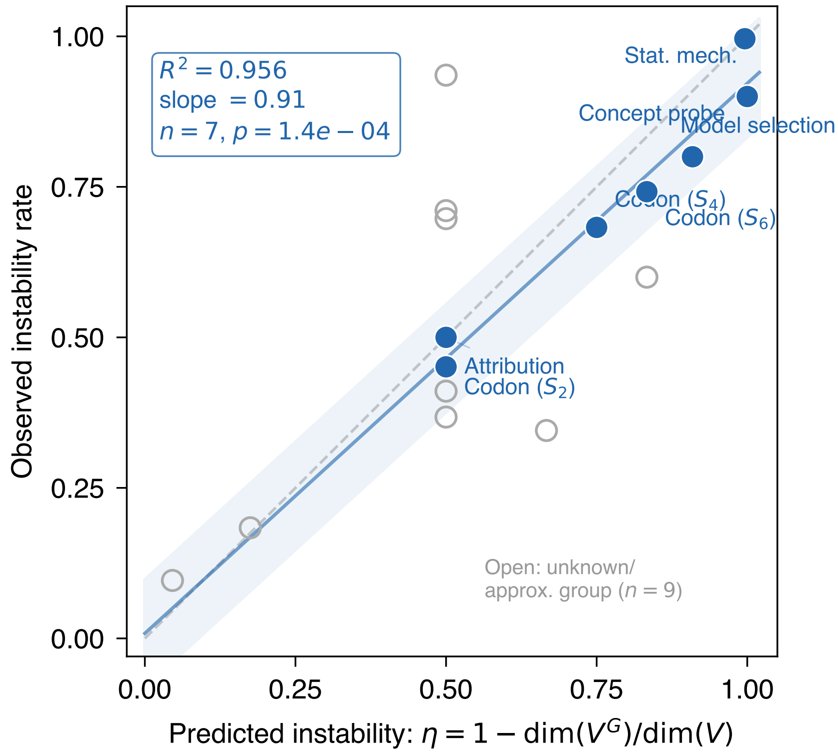

Figure 2: **Extended Data Figure 1.** The character-theoretic formula $\eta = 1 - \dim(V^G)/\dim(V)$ predicts observed instability across 16 domain instances. Filled circles: 7 instances with known symmetry groups ($R^2 = 0.957$, $p = 1.4 \times 10^{-4}$). Open circles: 9 instances with unknown or approximate groups. The bottleneck is group identification, not the law.

Code availability

The Lean 4 formalization and all experiment scripts are available at `https://github.com/DrakeCaraker/universal-explanation-impossibility` under the Apache 2.0 licence. The attribution-specific formalization is at `https://github.com/DrakeCaraker/dash-impossibility-lean`.

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Author contributions

D.C. conceived the framework, proved the theorems, wrote the Lean formalization, designed and executed the experiments, and wrote the manuscript. B.A. and D.R. contributed to experimental design, reviewed the proofs, and edited the manuscript.

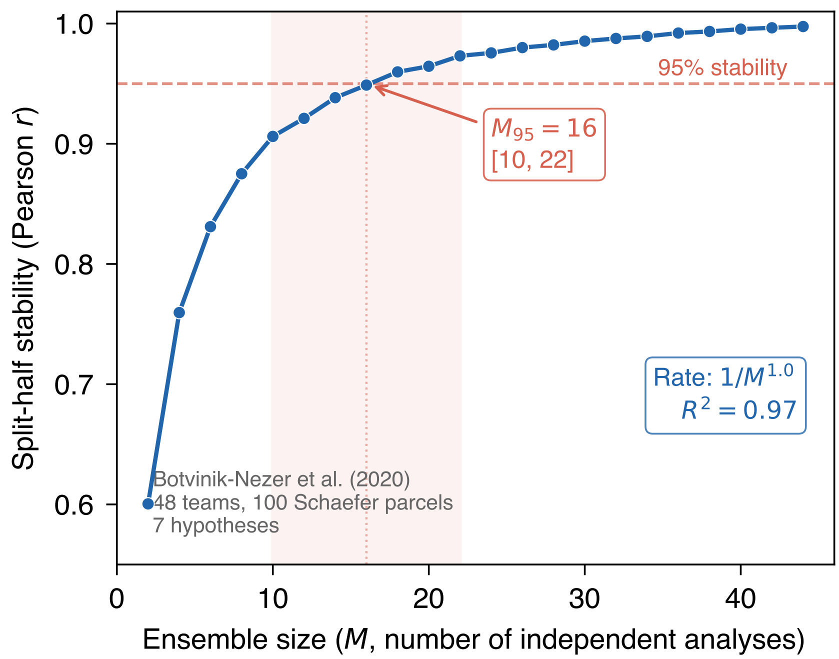

Figure 3: **Extended Data Figure 2.** Split-half stability of the ensemble average vs ensemble size for 48 neuroimaging analysis teams. Stability exceeds 0.95 at $M = 16$ [95% CI: 10–22]. Convergence rate: $1 - a/M^{1.0}$ ($R^2 = 0.97$).

Competing interests

The authors declare no competing interests. D.C.'s employment at Oura Health is unrelated to this work.

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Table 4: **Extended Data Table 2.** Twenty-three impossibility theorems from 14+ domains classified by tightness type. Full: every property pair achievable ("pick two" works). Collapsed: property pairs blocked (enrichment required). Structural: non-trivial Lean proof over unbounded solution space. Model: 3-element type that forces its tightness. Bell and Kochen-Specker tightness is decomposition-dependent (see text). The GL(n) result holds for all $n \geq 2$ and all primes p (Lean-verified).

System	**Domain**	**Tightness**	**Evidence**
Bilemma	**ML explanation**	**Collapsed**	**structural**
Quantum linearity	**Quantum info**	**Collapsed**	**model**
GL(n) representation	**Langlands program**	**Collapsed**	**structural**
Arrow	Social choice	Full	structural
Gibbard-Satterthwaite	Social choice	Full	structural
Bell (CHSH)	Quantum physics	Full	structural
Kochen-Specker	Quantum physics	Full	structural
Gödel incompleteness	Mathematical logic	Full	structural
ML fairness	ML prediction	Full	model
Bias-variance	Statistics	Full	model
Sen's liberal paradox	Social choice	Full	model
CAP theorem	Distributed systems	Full	model
FLP impossibility	Distributed systems	Full	model
Mundell-Fleming	Economics	Full	model
Perpetual motion	Thermodynamics	Full	model
Second law	Thermodynamics	Full	model
Münchhausen trilemma	Epistemology	Full	model
Competitive exclusion	Ecology	Full	model
Penrose-Hawking	General relativity	Full	model
Eastin-Knill	Quantum computing	p12-blocked	model
Shannon secrecy	Cryptography	p23-blocked	model
Navier–Stokes (3D)	Math. physics	Smooth-blocked	numerical (54 runs)
Navier–Stokes (2D)	Math. physics	Full (control)	numerical

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